An Algorithm for Modelling Rolling Intrinsic Battery Trading on the Continuous Intraday Market

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We introduce a novel rolling intrinsic algorithm to model battery-based trading on the continuous intraday market. Our approach leverages a discretization method with a subsequent step-wise optimization based on continuous price updates and complete uncertainty about future prices. We show the high potential of generating additional profits in the continuous intraday market in comparison to using an ID3-based representation of trading, which is frequently used in literature. Thereby, we realistically model actual trading behavior on the intraday market. We publish our code open-source to contribute to the further analysis of battery storage systems on the intraday market.

CCS Concepts: • **Computing methodologies** → *Model development and analysis.*

Additional Key Words and Phrases: battery trading, intraday market, continuous trading, energy markets

Availability of Data and Material:
The source code is available as an open-source Github repository.

1 INTRODUCTION

Battery storage systems (BESS) are playing an integral role in the clean energy transition to overcome the problem of intermittent solar and wind energy generation [Kittner et al. 2017]. To achieve a high proliferation of battery systems in modern power systems, developing viable business models to operate large-scale systems is integral. However, most studies find that the operation of BESS on wholesale markets is still unprofitable [Baumgarte et al. 2020]. One potential additional business model is "trading arbitrage" [Baumgarte et al. 2020], where the BESS is charged during low-price periods and discharged during high-price periods. Thereby, it also contributes to the smoothing of intermittent generation.

BESS can be operated on different markets, for instance, the auction-based day-ahead market or the continuous intraday market [Baumgarte et al. 2020; Braeuer et al. 2019]. The growing share of volatile generation from renewable energy requires market participants to balance their positions with shorter lead times and, hence, causes more traded volume on continuous intraday markets, as seen in Germany [Koch and Hirth 2019]. Internationally, many countries have implemented a continuous intraday market as well, making this a promising market for BESS participation [Scharff and Amelin 2016].

Studies that analyze BESS operations in the intraday market are typically based on a heavily simplified trading model of BESS, e.g. assuming that only one transaction every 15 minutes is executed [Braeuer et al. 2019], perfect knowledge about future prices is assumed [Cho and Kleit 2015] or potential purely financial shadow transactions, where the trades are later reversed, are neglected [Boukas et al. 2021]. These gaps in research lead to a potential underestimation of the generated profit of battery storage systems in the intraday market.

Hence, we provide an algorithm that extends the BESS trading options on the continuous intraday market. The proposed algorithm is easy to implement, corresponds to the continuous character of the market under uncertain future prices, and enables potential purely financial shadow trades, which lead to profits without requiring the battery to be discharged. The potential of purely financial, battery-backed transactions of the intraday market has not yet been discussed and analyzed in previous research. Overall, we thereby answer the following two research questions:

(1) How can continuous BESS trading on the intraday market be modeled with a rolling intrinsic model that considers uncertain future prices and possible financial countertrades?
(2) How high is the potential financial impact of the proposed rolling intrinsic intraday trading algorithm compared to an ID3 index-based benchmark model?

We answer these research questions with our study, which is structured as follows. First, we introduce related studies about intraday trading algorithms and develop the research gap at hand (Section 2). Then, in Section 3, we depict the German intraday market structure. Section 4 presents our methodology, consisting of the discretization of trades and a rolling optimization model. In addition, we introduce an ID3 benchmark model and relevant metrics. In Section 5, we present our results, which are discussed in Section 6. Finally, in Section 7, we conclude the study and provide an outlook on further research questions.

2 RELATED WORK

Trading in the electricity market is increasingly complex due to ongoing market evolutions. This complexity is evidenced by a spectrum of methods and varied problem scopes that are examined within the ambit of trading electricity studies. In contrast to day-ahead market modeling, literature that deals with the continuous intraday markets is scarce [Glas et al. 2020]. An extensive array of

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literature does exist that analyzes the price movements and predictability aspects of intraday trading, as documented in sources such as [Hagemann 2013; Kiesel and Paraschiv 2017; Narajewski and Ziel 2020; Pape et al. 2016]. While these studies shed light on market determinants, they refrain from delving into the practical trading strategies in the continuous markets [Kath and Ziel 2020].

The methodologies employed to assess trading strategies in the intraday market vary widely, often necessitating an apriori quantification of underlying price movements. [Braun and Hoffmann 2016] present a case study based on a Mixed Integer Problem modeling approach for a pumped hydro storage system that considers both the continuous intraday and day-ahead markets in Germany. They focus mainly on technical constraints related to the storage system. [Kraft et al. 2023] formulate a Multi-Stage Stochastic Mixed Integer Linear Program for a portfolio consisting of biogas and photovoltaic assets. Their trading strategy considers day-ahead, balancing, and continuous intraday markets, as well as long and short positions exceeding the capacity of the underlying asset. [Ottesen et al. 2018] formulate an intraday trading strategy for an aggregator of demand flexibility. [Finnah et al. 2022] facilitate an Approximate Dynamic Program for the bidding problem of an energy storage in the day-ahead and intraday markets.

While the foregoing studies consider the electricity market interrelations, they simplify the continuity of the intraday market. This means that each intraday market product is tradeable with one auction where a volume-weighted price index is assumed to be realized, like the EPEX SPOT ID3. Such a simplification is suitable for a first approximation but neglects the trading character of the intraday market by reducing the dimension of different trading times [Kath and Ziel 2020]. In real-world scenarios, the continuous intraday market has frequent and sometimes rapid price changes due to various factors such as supply and demand imbalances, especially due to forecasting errors of intermittent RES [Kiesel and Paraschiv 2017]. These intraday price differentials allow market participants to re-optimize existing positions based on new prices, such as modeled by [Boukas et al. 2021], potentially leading to increased revenue for the BESS.

Some studies also explore the concept of continuity in trading strategies. We give an overview of exemplary studies in Table 1. Most notable, [Gönsch and Hassler 2016], [Bertrand and Papavasiliou 2019] and [Boukas et al. 2021] propose a trading algorithm that employs a Markov Decision Process MDP framework to identify optimal policies. [Bertrand and Papavasiliou 2019] and [Boukas et al. 2021] also provide a rolling intrinsic algorithm that simulates continuous trading of energy storage, which considers trading decisions based on the available information at each time step. Both studies show the advantages of an MDP consideration compared to a rolling intrinsic policy. For example, in [Bertrand and Papavasiliou 2019], the rolling intrinsic policy only considers buying power if it can sell it directly. This is because the method maximizes the profit of the current time step and ignores future trading. [Hassler 2017] also model the continuous intraday trading as an MDP, yet they derive applicable decision rules from the found solution for a real-world application. These rules performed well compared to a sophisticated Approximate Dynamic Programming algorithm adapted from literature. [Bertrand and Papavasiliou 2019] and [Hassler 2017] showcase the suitability of an accurate depiction of the continuity of the intraday auctions and bridge the MDP results to the real-world application. We build upon this narrative and propose an algorithm that also considers future trading possibilities in continuous markets while using application-ready optimization. Like this, we can omit the necessary steps to parameterize the price thresholds and optimize the resulting policy functions as in [Bertrand and Papavasiliou 2019].

We base the optimization on [Martin and Otterson 2018], which provides a model for the EPEX SPOT German continuous intraday market based on limited order book LOB data. They discuss possible trading decision rules and necessary preprocessing steps for EPEX SPOT market data. According to them, the proposed market model of the intraday market should be used to simulate intraday trading of flexible market participants to increase realism. The application of this model facilitates a detailed analysis of order execution within continuous intraday markets, an area that is in need of further investigation [Kath and Ziel 2020]. Given its growing importance in the electricity system, we selected a BESS as the underlying asset for our analysis. BESS are particularly compelling assets for intraday trading as they allow for the possibility to procure power from relatively cheap sell bids, store the power, and sell it back to subsequent buy bids that place a greater valuation on the procured power [Bertrand and Papavasiliou 2019].

With this approach, we fill the current research gap regarding application-oriented arbitrage trading using an underlying asset such as BESS in the continuous intraday market. We analyze the advantage of explicitly leveraging the continuity of the market and financial transactions by comparing an ID3 price simplification [Kraft et al. 2023; Ottesen et al. 2018] with our approach. To further deepen the application focus of the proposed algorithm, we consider the most important lithium-ion BESS specifics, such as cyclic warranty conditions. To assess the relevance of these constraints, we conduct a sensitivity analysis that examines the effect of adjusting these constraints on gross profits.

In addition, studies considering trading strategies for intraday auctions are based on data from 2011 to 2016 [Bertrand and Papavasiliou 2019; Boukas et al. 2021; Metz and Saraiva 2018]. Based on this data [Metz and Saraiva 2018] find that even arbitrage profits are not enough to justify the investment cost of BESS. To further accelerate these discussions and the proliferation of BESS we provide the proposed algorithm open-source. Moreover, we explore more recent market data (2022) for the evaluation of the trading strategies. As the market dynamics and prices have changed significantly in recent years, our analysis offers a timely and updated understanding of the current market landscape and a starting point for upcoming profitability discussions.

3 CONTINUOUS INTRADAY MARKET

Even though an intraday market is present in multiple electricity systems around the world [Cho and Kleit 2015] and in Europe mostly harmonized, some differences remain. In the following, we provide an overview of the German market structure and specific lead times. Yet, the proposed approach can be implemented in various continuous intraday markets.

Paper	Assets	Markets	Data	Model
[Bertrand and Papavasiliou 2019]	PHES	CI	LOB, DE/LU (2015, 2016)	MDP
[Boukas et al. 2021]	PHES	CI	LOB, DE/LU (2015)	MDP
[Braun and Hoffmann 2016]	PHES	DA, CI	LOB/Auctions, DE/LU (2015)	MILP
[Metz and Saraiva 2018]	Li-Ion BESS	IA	Auctions, DE/LU (2015)	MILP
[Braun 2016]	PHES	DA, IA	Auctions, DE/LU (2015)	MILP
[Kraft et al. 2023]	Biogas, PV	DA, FR, IA	ID3/Auctions, DE/LU (2019)	Stoch. MILP
[Ottesen et al. 2018]	Flexible Demand	DA, FR, IA	ID3/Auctions, NO (2019)	Stoch. MILP
[Gönsch and Hassler 2016]	Wind, Li-Ion BESS	CI	LOB, DE/LU (2013)	MDP/ADP
[Hassler 2017]	Wind, Li-Ion BESS	CI	LOB, DE/LU (2014)	MDP/ADP
[Löhndorf and Wozabal 2022]	PHES, BESS	DA, CI	LOB, DE/LU (2017)	Stoch. MILP

Abbreviations: PHES - Pumped Hydro Energy Storage, CI - Continuous Intraday, DA - Day-Ahead, IA - Intraday Auction, FR - Frequency Restoration, LOB - Limit Order Book, DE/LU - Germany/Luxembourg, NO - Norway, MDP - Markov Decision Process, MILP - Mixed-Integer Linear Programming, Stoch. MILP - Stochastic Mixed-Integer Linear Programming, ADP - Approximate Dynamic Programming

Table 1. Overview of related literature for trading in the intraday market

3.1 German case study

The main electricity intraday markets in Germany are operated by EPEX SPOT SE. They can be divided into two sub-markets: Intraday auctions of quarter-hourly products and the continuous intraday market [EPEX SPOT SE 2023].

In Germany, the intraday auctions occur at 15:00 CET for the quarter-hourly products of the following day. The higher resolution of 15 minutes allows market participants to adjust their positions more precisely and respond to smaller fluctuations in supply and demand [Baule and Naumann 2021].

In the continuous intraday market, market participants can continuously trade electricity throughout the day until shortly before delivery. This market operates in real-time, enabling participants to react to changing market conditions, unforeseen events, and new information that may impact their trading strategies. The continuous trading starts at 15:00 CET for Germany, with all 24 hours, 48 half hours, and 96 quarter hours of the following day as available products. [EPEX SPOT SE 2023]

Based on the area in which the matched trades are realized, the product lead times differ in the following way [EPEX SPOT SE 2023]:

(1) **Up to 1 hour before delivery:** EU cross-zonal intraday trading based on Single Intraday Coupling (*SIDC*)
(2) **Up to 30 minutes before delivery:** Cross-zonal trading within Germany
(3) **Up to 5 minutes before delivery:** Single delivery area trading (*SDAT*)

The continuous intraday market's structure differs vastly from the other auction-based electricity markets. Market participants have access to a LOB, as visualized in Figure 1), where they can submit their buy and sell orders for available products. The order book represents market participants' collective supply and demand, displaying the prices and volumes at which they are willing to trade. Market participants can adjust their orders quickly, modifying prices or volumes to reflect changing market conditions. This flexibility enables them to respond quickly to new information or events that may impact the supply and demand of the electricity market. By

Fig. 1. Example of a limited order book for a specific product at a specific time

actively managing their orders, participants can re-optimize their positions and benefit from trading opportunities as prices deviate throughout the day [Martin and Otterson 2018].

The LOB nature of the continuous intraday market means that market participants can observe the existing orders and their corresponding prices and volumes before deciding to execute their own trades. This is a feature not present in auction-based electricity markets.

Operators of electricity storage systems can leverage the continuous intraday market by capitalizing on price differentials and market fluctuations. Since all 24 hours (as well as all 48 half-hours and 96 quarter-hours) are traded simultaneously, there are risk-free arbitrage opportunities during the day, where an operator can buy a specific hourly product at a relatively low price and sell another, later hourly product at a higher price simultaneously [Boukas et al. 2021].

As outlined in this Section, the motivation for the detailed examination of trading models for the continuous intraday market in this study is based on the increasing relevance of the market [EPEX SPOT SE 2022b], as well as the additional revenue potential

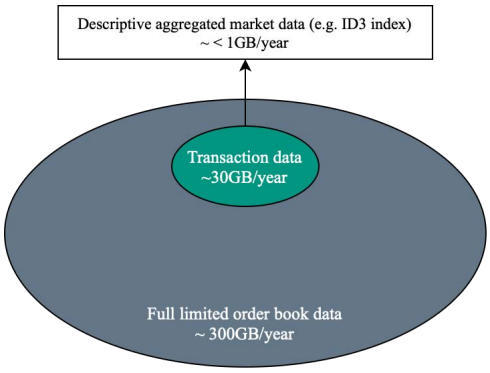

Fig. 2. Data sets for continuous intraday trading at EPEX SPOT (2022)

arising from the LOB format of the market, which results in price fluctuations during the day that flexibility providers can exploit.

3.2 Data structure

Data about past transactions on the continuous intraday market are available in various granularities. Understanding the underlying data structure is essential for building algorithms based on the LOB and reflecting daily market price fluctuations.

In Figure 2, we provide an overview of data sets of different granularity and detail available for the EPEX SPOT intraday market [Martin and Otterson 2018]. EPEX SPOT operates electricity markets in many European countries, running day-ahead and capacity auctions, as well as operating the continuous intraday market. In the following, we introduce the different data sets provided by EPEX SPOT post-trading for its intraday market:

(1) **Aggregated market data**: EPEX SPOT provides free aggregated market information for continuous markets. This data includes minimum and maximum prices for the available contracts, volume-weighted averages, and information about the traded volume for each contract. In addition, indices are defined to provide low-granularity information about the products traded: The ID1 index summarizes price information for the products by calculating a volume-weighted average price of overall transactions executed in the last hour before delivery. Respectively, the ID3 index calculates a volume-weighted average price over all transactions executed in the last three hours before delivery, and the IDFULL index provides a volume-weighted average over all transactions for a specific product. [Martin and Otterson 2018, 3]

(2) **M7 Order book data**: EPEX SPOT offers complete historical order book information for the continuous intraday market as a paid service. The order book offers an in-depth view of market dynamics and can be used to inform trading strategies and conduct detailed market analysis. The order book data provided by EPEX SPOT includes millions of orders processed daily. This includes both the orders that actually resulted in transactions and limit orders that never led to an execution. [Martin and Otterson 2018, 3]

(3) **Transaction data**: Besides full order book data, EPEX SPOT provides a subset of the order book that only includes orders that resulted in transactions (roughly one-tenth the size of full order book data). [Martin and Otterson 2018, 3]

While aggregated market data exhibit less intraday trading volatility due to the averaging over longer time-horizons, both full order book and transaction data could be used for trading simulation with shorter time steps. Full order book data is the most detailed and realistic market representation, capturing the depth and dynamics of trading activity. However, for the purpose of the presented trading simulation, we argue for employing the transaction data set for the following reasons [Rominger et al. 2019]. Transaction data provides a clear lens into the actual market outcomes — the prices and quantities where demand intersects with supply. The complete nature of order book data can also lead to noise in the data set [Martin and Otterson 2018]. Orders, especially those deep in the book and distant from realized prices, might be more speculative than genuine. By using only transaction data, the simulation focuses solely on confirmed, executed trades. While the full order book data is the most thorough representation of trades, it also introduces a significant computational overhead, as shown by the approximate data set sizes in Figure 2. Concentrating on transaction data simplifies the simulation process, eliminating the potential noise without compromising essential insights. While processing and storing the full order book data was not feasible with the resources available for this work, the transaction data subset provides a reasonable compromise between sufficient data granularity for estimating the results of the trading strategies discussed and the resources needed to process and store the data.

For these reasons, we use the transaction data set as the base data set for our rolling intrinsic trading simulation, which is described in the following chapter.

4 METHODOLOGY

In this Section, we provide an overview of our method to model BESS trading on the continuous intraday market based on the transaction data set. In the first step, we discretize the observations from the transaction data. Then, given past trades and new transaction data, we make trading decisions during every discretized time step. We describe our methodology in the following in detail.

4.1 Discretization

Successful trades on the continuous intraday market occur irregularly, as orders are matched continuously. For our approach, the transaction data is hence grouped into discrete intervals. The applied discretization methodology and the following notation are derived from [Kath and Ziel 2020]. Through discretization, the computational complexity of the intraday trading task is reduced, while a realistic representation of continuous orders is still maintained. In comparison, considering every available buy or sell order, at least one simulation step per second would be required. In the case of 5-minute discretization intervals, a second-wise optimization would lead to a 300-fold increase of time steps in the trading optimization. Hence, for the sake of computational efficiency, our study employs a bucket-wise discretization approach. We note that in future studies,

also a consideration of all underlying orders could be an interesting benchmark scenario.

The transaction data can be formalized as a set Z. The execution times of the given transactions are defined as a set T such that $T \subseteq [t_{\text{start}}, t_{\text{end}}]$, where $[t_{\text{start}}, t_{\text{end}}]$ marks the trading period to be simulated:

$$Z = \{(y_t, v_t, p_t, d_t) | t \in T, T \subseteq [t_{\text{start}}, t_{\text{end}}]\} \tag{1}$$

The set Z is comprised of the trades at execution time t with the execution price of the trade denoted as $y_t \in \mathbb{R}$, the volume of the trade as $v_t \in \mathbb{R} | v_t > 0$, the product traded (e.g. delivery between 12:00-12:15) as $p_t \in P$, where P is the set of all tradable products, and the direction of the trade as $d_t \in buy, sell$. The set of all tradable products could, for instance, be all 15-minute or 1-hour delivery periods per day, depending on the market at hand.

With t_{start} being the start of the trading period, a bucket function $B(t)$ can be defined given a bucket size b:

$$B(t) = \left\lfloor \frac{t - t_{\text{start}}}{b} \right\rfloor \tag{2}$$

Based on the bucket size b and the trading period $[t_{\text{start}}, t_{\text{end}}]$ discrete time buckets can be defined as O_k:

$$O_k = \{(t, y_t, v_t, p_t, d_t) \mid t \in T, B(t) = k\} \tag{3}$$

This methodology splits the transaction data into equidistant time buckets indexed by k, where $k \in \left[0, \left\lfloor \frac{t_{\text{end}} - t_{\text{min}}}{b} \right\rfloor\right]$.

For every time bucket k (e.g., the first 5-minute time span of the simulation horizon), product p and trade direction d, a volume-weighted average price $\text{VWAP}_{k,p,d}$ can be defined as:

$$\text{VWAP}_{k,p,d} = \frac{\sum_{t \in T_k, p_t = p, d_t = d} y_t \cdot v_t}{\sum_{t \in T_k, p_t = p, d_t = d} v_t} \tag{4}$$

Due to the nature of the transaction data, where every buy trade matches a respective sell trade with the same volume and price, volume-weighted average prices are equal for both trading directions:

$$\text{VWAP}^{\text{buy}}_{k,p} = \text{VWAP}^{\text{sell}}_{k,p} \tag{5}$$

Based on Equation 5, Equation 4 can be simplified to:

$$\text{VWAP}_{k,p} = \frac{\sum_{t \in T_k, p_t = p} y_t \cdot v_t}{\sum_{t \in T_k, p_t = p} v_t} \tag{6}$$

We, therefore, simplify the trading reality in a continuous LOB market where spreads between bid and ask orders, as visualized earlier in Figure 1, must be considered.

For every time bucket k, a volume-weighted average can be defined. With the following equation, we only include time buckets with at least 10 transactions for a given product:

$$\text{VWAP}^{\text{valid}}_{k} = \left\{\text{VWAP}_{p,k} \mid W(p, O_k) > 10\right\} \tag{7}$$

where $W(p, O_k)$ is the number of transactions in the subset O_k for product p. The constraint $W(p, O_k) > 10$ is introduced for two reasons. First, a product that has been traded more frequently within a given time bucket generally signifies that there's sufficient market interest or liquidity for that product. This is important when calculating a metric like VWAP because, for products with very few trades, the VWAP can be highly sensitive to individual trades. By ensuring a minimum number of transactions for calculating the VWAP, we ensure that the VWAP represents a broad and realistic market consensus for a given product at a specific time bucket. Especially in the hours far away from the delivery time, fewer trades are made [Löhndorf and Wozabal 2022]. Occasional trades might occur at prices that deviate significantly from the norm for various reasons (e.g., input errors, strategic trades, etc.). Such trades can considerably skew metrics like the VWAP, especially if the volume associated with the outlier price is high. A recent extreme example of this can be seen in the transaction data for 18.09.2023 for the DE/LU market area, where the delivery period 12:00 to 12:15 was traded at approx. 190€/MWh before two distinct transactions were executed for a price of 6999.98 EUR/MWh. Prices returned to normal immediately afterward. By only considering products with a higher number of trades, the impact of such outliers on trading simulation results is reduced significantly. By setting $W(p, O_k) > 10$, we ensure a reasonable trade-off between realistic market consensus while still having enough tradable buckets. In addition, we provide a sensitivity analysis of minimum trades per bucket and resulting gross profit in the Appendix.

4.2 Rolling intrinsic trading model

In the following, we describe our rolling intrinsic trading algorithm, based on the discretized transaction data. The algorithm approximates continuous intraday trading by iterating over time buckets and making trading decisions while ensuring constraints are met, as described in Algorithm 1.

Algorithm 1 Battery Trading Simulation

1: **Initialization**: Battery and trading parameters are initialized. These parameter sets are provided as inputs to the simulation.
2: **for** $k = 0$ **to** K **do** ▹ Iterate through each time period, from $k = 0$ to $k = K$
3: Retrieve the set $VWAP^{\text{valid}}_k$ of tradable products in the given time period k.
4: Identify the most profitable combination of potential trades in the period k, given tradable products $VWAP^{\text{valid}}_k$, based on an optimization model
5: Apended these trades to the list of previous trades, serving as inputs for subsequent periods.
6: **end for**
7: **Output**: The set of trades is returned and can be used to calculate metrics such as full load cycles or gross revenues generated by trading.

The rolling intrinsic framework was first presented in the domain of gas storage valuation [Bjerksund et al. 2011; de Jong 2015; Gray and Khandelwal 2004]. The idea of the rolling intrinsic approach is to perform multiple optimization runs in a given frequency over the course of the trading period. This enables the storage operator to re-optimize the positions of previous optimization runs given new price information. The rolling intrinsic model implemented in

this study is based on this basic idea and adopted for the use case of an electricity storage operator in the continuous intraday market [Boukas et al. 2021].

A mixed integer optimization model is formulated to find profit-maximizing trades at any point in time (in our case, within a time bucket).

Therefore, we introduce the following variables:

- **Buy Amount ($b_{k,p}$)**: This variable denotes the amount of energy bought in period p within the optimization run in time bucket k. It is measured in megawatt-hours (MWh).
- **Sell Amount ($s_{k,p}$)**: This indicates the quantity of energy sold from the battery in period p within the optimization run in time bucket k, also measured in MWh.
- **State of Charge ($\mathbf{SOC}_p$)**: This variable reflects the battery's state of charge (SOC) at the start of period p, also in MWh.

Several parameters for the battery operations are implemented:

- **Round-trip Efficiency (η)**: This represents the combined efficiency of the charging and discharging processes, as typically used as a metric for life-cycle evaluations of BESS [Hiremath et al. 2015].
- **Battery Capacity (C)**: This is the maximum energy content the battery can store in MWh.
- **C-Rate (r)**: This represents the maximum rate at which the battery can charge or discharge as a percentage of the maximum battery capacity. The C-Rate can be formalized as $r = \frac{I}{C}$, with I being the charge–discharge current of the system [Collath et al. 2022, 5]. The choice of r is a critical decision, as high charging and discharging rates lead to increased cyclic aging [Naumann et al. 2020].
- **Total Cycles allowed per day (N)**: This indicates the total number of charging cycles the battery undergoes in a day. This parameter will simulate warranty conditions, which often include a maximum number of cycles per time period [Fares and Webber 2018; Kelly and Leahy 2020; Mongird et al. 2019; Xu 2022].

A trading algorithm can, therefore, be formulated as follows for a specific time bucket k. The optimization goal is maximizing the profits at each discretized bucket k. The profits are constituted by the sum over all tradable products p of sold quantities $s_{k,p}$, deducted by bought quantities $b_{k,p}$ at the respective prices $\mathrm{VWAP}_{k,p}$:

$$\max \sum_p \left(s_{k,p} \cdot \mathrm{VWAP}_{k,p} - b_{k,p} \cdot \mathrm{VWAP}_{k,p} \right) \tag{8}$$

The tradable product p can also be interpreted as the time step when the battery has to be charged or discharged. Various constraints have to be accommodated for realistically modeling continuous trading on the intraday market:

$$\mathrm{SOC}_1 = 0 \tag{9}$$

$$\mathrm{SOC}_p = \mathrm{SOC}_{k,p-1} + b_{k,p-1} \cdot \sqrt{\eta} - s_{k,p-1} \cdot \frac{1}{\sqrt{\eta}}, \quad \forall p > 1 \tag{10}$$

$$\mathrm{SOC}_p \leq C, \quad \forall p \tag{11}$$

$$b_{k,p} \leq C \cdot r, \quad \forall p \tag{12}$$

$$s_{k,p} \leq C \cdot r, \quad \forall p \tag{13}$$

$$s_{k,p} \cdot \frac{1}{\sqrt{\eta}} \leq \mathrm{SOC}_p, \quad \forall p \tag{14}$$

$$\sum_p b_{k,p} \cdot \sqrt{\eta} \leq N \times C \tag{15}$$

$$b_{k,p}, s_{k,p}, \mathrm{SOC}_p \geq 0, \quad \forall p \tag{16}$$

In Equation 9, the initial SOC of the battery is set at 0, Equation 10 controls the $SOC_{k,p}$ based on buy and sell decisions, Equations 11 - 13 enforce that the battery is operated within technical constraints. Equation 14 and 15 consider efficiency losses, Equation 16 enforces the non-negativity of $SOC_{k,p}$, the buy and sell quantities $b_{k,p}$ and $s_{k,p}$.

This algorithm returns trades for any given list of products and their respective prices ($VWAP_p$) while ensuring that technical restrictions are met. Because of the continuous trading format, this base algorithm has to be extended to account for trades made in previous time buckets.

This is achieved by introducing three more variables into the optimization problem:

- **Net Buy Amount ($\tilde{b}_p$)**: This is the quantity that is bought for one product after taking into account all previous trades as well as the trades within the current optimization run.
- **Net Sell amount ($\tilde{s}_p$)**: This is the quantity that is sold for one product after taking into account all previous trades as well as the trades within the current optimization run.
- **Charge Sign (λ_p)**: This is a binary variable, which is 1 if the battery is charging for a specific time period (or product) and 0 if it is discharging.

Following this modeling approach, the constraints affecting maximum net buy and sell amounts have to be adjusted based on the binary charge sign λ_p:

$$\tilde{b}_{k,p} \leq C \cdot r \times \lambda_p, \forall p \tag{17}$$

$$\tilde{s}_{k,p} \leq C \cdot r \times (1 - \lambda_p), \forall p \tag{18}$$

In addition, the following constraint is added:

$$\tilde{b}_{k,p} \times \lambda_p - \tilde{s}_{k,p} \times (1 - \lambda_p) = b_{k,p} + \sum_{i=1}^{k-1} b_{i,p} - s_{k,p} + \sum_{i=1}^{k-1} s_{i,p}, \forall p \tag{19}$$

While Equation 17 and Equation 18 are used to determine the current charge sign (whether the battery is charging or discharging), Equation 19 calculates the actual net charging or discharging amounts for each product (or delivery period) p. The differentiation between charging and discharging amounts is necessary to account

for inefficiencies in the battery correctly: When calculating the battery SOC, net buy amounts have to be multiplied by the efficiency factor $\sqrt{\eta}$ (as not all energy bought makes it to the battery) and net sell amounts have to be divided by $\sqrt{\eta}$ (as more energy has to be extracted from the battery than was sold).

Overall, we replace in our optimization model the sell and buy amounts with the now introduced net amounts to consider previous trades:

$$\tilde{s}_{k,p} = s_{k,p} \tilde{b}_{k,p} = b_{k,p} \tag{20}$$

The introduced model now optimizes the positions without relying on forecast information by simultaneously capitalizing on the possibility of arbitrage between various contracts. It is important to highlight that the rolling intrinsic algorithm allows for repositioning. Unlike traditional strategies for auction-based markets, which can only hold a position based on an initial decision, this algorithm re-evaluates and might buy or sell back previous positions throughout the day. This dynamic approach ensures that the battery operates at its optimal state, taking advantage of the volatility in the intraday market. Furthermore, an important aspect of this approach is the occurrence of only financial trades. These are trades executed by the algorithm that do not necessarily lead to actual battery discharging or charging. Instead, they provide additional revenue potential by adjusting the battery's market position and capitalizing on price fluctuations throughout the trading period.

4.3 Benchmark: ID3

We benchmark our rolling intrinsic algorithm against a frequently used approach in research, that utilizes the ID3 price index to approximate continuous intraday trading [Kraft et al. 2023; Ottesen et al. 2018]. With this approach, studies, for example, attempt to depict the general option of repositioning on the intraday market within a multi-market setting. While aggregating prices on a higher level, the index gives a descriptive overview of the prices of the continuous markets and is therefore included as a benchmark for the evaluation. With this, we can assess the increased monetary benefit of the rolling intrinsic algorithm on the continuous intraday results in comparison to ID3 data.

In contrast to the previously described trading simulation with transaction data, the ID3 simulation is solved using a single optimization problem under perfect foresight, making trade, charge and discharge decisions based on the underlying ID3 price index. In contrast, the previously introduced rolling intrinsic algorithm has no perfect foresight, thereby making actual operations on the continuous market more realistic. The overall optimization goal in the benchmark approach is to maximize the difference between sold quantities s_p at price $\text{VWAP}_p^{\text{ID3}}$ and the bought quantities b_p at price $\text{VWAP}_p^{\text{ID3}}$ for each time step p:

$$\text{maximize} \quad \sum_p \left(s_p \cdot \text{VWAP}_p^{\text{ID3}} - b_p \cdot \text{VWAP}_p^{\text{ID3}} \right) \tag{21}$$

The optimization is subject to the initial technical constraints from Equation 10 to 16.

4.4 Performance metrics

We compare the rolling intrinsic and ID3-based simulation approach based on two metrics introduced in the following: gross trading profits and profits per cycle.

Gross Trading Profit (π)

Gross profit represents the total monetary gain from sales of energy while accounting for the direct costs of obtaining the energy but not accounting any other costs. Specifically, trading costs and wear on the battery are not considered.

The metric offers a standardized way of comparing various trading algorithms, focusing on their core performance without the complexities of individual cost structures. Given the adherence to a given number of full load cycles per year, battery wear concerns are primarily mitigated by warranty conditions [Fares and Webber 2018; Kelly and Leahy 2020].

The gross trading profit for the static priced ID3 benchmark model is given by the subtraction of all sold from the bought quantities multiplied by their prices:

$$\pi = \sum_{p \in P} \left(s_p \cdot \text{VWAP}_p^{\text{ID3}} - b_p \cdot \text{VWAP}_p^{\text{ID3}} \right) \tag{22}$$

The gross trading profit for the rolling intrinsic model is given by the sum of gross quantities bought and sold over all time buckets in the trading period weighted by their respective $VWAP$:

$$\pi = \sum_{k \in K} \sum_{p \in P_k} \left(s_p \cdot \text{VWAP}_{k,p} - b_{k,p} \cdot \text{VWAP}_{k,p} \right) \tag{23}$$

Profit Per Cycle (PPC)

While our study neglects a detailed degradation analysis of the simulated storage system, we assume that a certain number of load cycles should not be exceeded due to warranty conditions [Ahmadi et al. 2017; Viswanathan et al. 2022]. By introducing the Profit per Cycle (PPC) metric, a more nuanced understanding of a trading strategy's efficiency by relating the generated cash flow to the number of full load cycles is enabled.

As lithium-ion BESS are examined in this study, this metric is introduced to ensure that revenues are not simply examined on an absolute scale but are contextualized against the approximate wear they impose on the BESS.

The PPC can be formulated as:

$$PPC = \frac{\pi}{\text{N}} \tag{24}$$

where N is the number of full load cycles used by the trading strategies. In our case, for the static priced ID3 benchmark model this can be calculated by:

$$N = \sum_{p \in P} b_p \times \sqrt{\eta} \tag{25}$$

In the case of the rolling intrinsic model the cycles used can be calculated by:

$$N = \sum_{k \in K} \sum_{p \in P_k} \tilde{b}_{k,p} \times \sqrt{\eta} \tag{26}$$

We apply these metrics in the following on a year-long case study.

5 RESULTS

This Section applies the proposed rolling intrinsic trading simulation algorithm to a year-long case study. We then compare the results with an ID3 index-based simulation frequently used in energy market trading studies [Kraft et al. 2023]. In the following, the rolling intrinsic models are abbreviated with *RI CONT*, while the ID3 models are denoted as *PF ID*3. For our case study, we use transaction data from the German continuous intraday market from 2022 [EPEX SPOT SE 2022a] for the quarterly hour products (15m) and hourly products (60m). Although our study utilizes German market data, the approach could easily be implemented on other international intraday electricity markets [Scharff and Amelin 2016].

We assume a battery capacity (C) of 1MWh, based on current large-scale projects [Petrova 2023], a round-trip efficiency (η) of 0.86 [Viswanathan et al. 2022] and one allowed equivalent full cycle per day, leading to 365 allowed warranty cycles (N) per year, based on [Komorowska et al. 2022]. The bucket size (b) is set at 15 minutes.

In Figure 3, we illustrate the rolling intrinsic simulation over one day for the hourly products based on the volume-weighted average prices ($\text{VWAP}_{k,p}$). Over the day, some hourly products are traded multiple times, thereby realizing financial revenues that are not necessarily associated with battery charging and discharging cycles. For example on the shown day the product for hour 13 p.m. to 14 p.m. is traded 32 times in total. This is a result of the underlying uncertainty in the price movement that we are considering. The reoptimisation in the rolling intrinsic optimisation facilitates new price movements as they occur. Ultimately, the battery is charged at lower prices mid-day and discharged during price peaks in the evening.

An overview over the behavior of the benchmark optimization on ID3 for an exemplary day is presented in Figure 4. Here, only one transaction per product is performed. Since no repositioning in the market is modeled here, all trades result in charging and discharging events of the battery. For the chosen exemplary day, this results in two full charging cycles. The *PFID*3 simulation facilitates the two largest price spreads and entirely charges/discharges the battery twice. Although we also restrict the yearly battery cycles in the *PFID*3 optimization, two cycles can be observed on the depicted day. This is due to the perfect foresight optimization over the whole year, thereby not operating the BESS on another, less profitable day. Contrary, the *RICONT* trading results in only one full charging cycle. These differences can also be analysed with the introduced metrics.

Model	π (€)	PPC (Eur / Cycle)
RI CONT (15m)	**111631.86**	**305.84**
RI CONT (60m)	70749.62	193.83
PF ID3 (15m)	80244.57	219.85
PF ID3 (60m)	57731.58	158.17

Table 2. Performance metrics of trading model and full-foresight benchmarks

We compare the metrics of our battery-based continuous intraday trading simulation in Table 2. In the difference of *PFID*3 and *RICONT*, we can observe that active repositioning in the continuous intraday market increases the profit potential of the battery by 39.2%. For both simulation models, the 15min products yield remarkably higher profits than the 60min products. This might be because the 15-minute products are used to adjust the hourly power outputs in the day-ahead markets. The time interval leads during a ramping up of production to an underproduction of power in the first two-quarter hours and an overproduction in the second two-quarter hours. This systematic repositioning of quarter-hour products leads to price fluctuations that can be facilitated by storage. Furthermore, we can see that the higher profits of the rolling intrinsic simulation *RICONT* are not caused by more cycles. Contrary even the profit per cycle (*PPC*) is higher for both product durations than for the ID3 counterpart. We see the ability for financial repositioning and the utilization of price fluctuations over the day as the main reason behind the higher profits exhibited by the *RICONT* model.

In Figure 5, the resulting cumulative profits over the two investigated product types (15min and 60min) and two models are depicted. The illustration shows that the steepest increase in cumulative profits can be observed over all methods during summer months, utilizing heavy price fluctuations caused for example by high fluctuating photovoltaic infeed.

In the Appendix, we run a sensitivity analysis for the most important parameters of the rolling intrinsic simulation to assess their impact on the presented results. We find that the smaller the bucket size, the higher the profits from the simulation. One reason for this is that smaller bucket sizes lead to more daily trading steps, thereby more trading and repositioning opportunities. Especially for the 15min rolling intrinsic algorithm, a higher number of allowed equivalent full cycles per year leads to substantially increased profits.

6 DISCUSSION

The generated trading trajectory by the proposed algorithm generates notably higher profits than simplifying the continuity with the ID3 prices despite assuming full uncertainty. First, we show that a significant battery profit potential lies in purely financial trades that are later resolved through counter trades. Thereby, it underlines the drawback of a simplification of the time dimension ID3 index-based simulation since repositioning and buying positions back is not possible, as well as a consideration of frequent trades, which can be represented through bucket discretization. Another benefit of our approach is that no perfect foresight of prices is assumed. Instead, trading decisions are based on continuous price updates, which could also be observed in a real-world scenario. As shown in the results, the re-optimisation in every step of the rolling intrinsic algorithm leads to a high amount of trades in total. We note that for a detailed profitability cost of continuous BESS trading, transaction costs and spreads should be considered - an analysis not often made in literature [Kath and Ziel 2020]. [Narajewski and Ziel 2022] assume transactions costs in the day-ahead market of 0.05 €/MWh and in the intraday start auction of 0.10 €/MWh. Such values are easily implementable in our algorithm with a threshold of

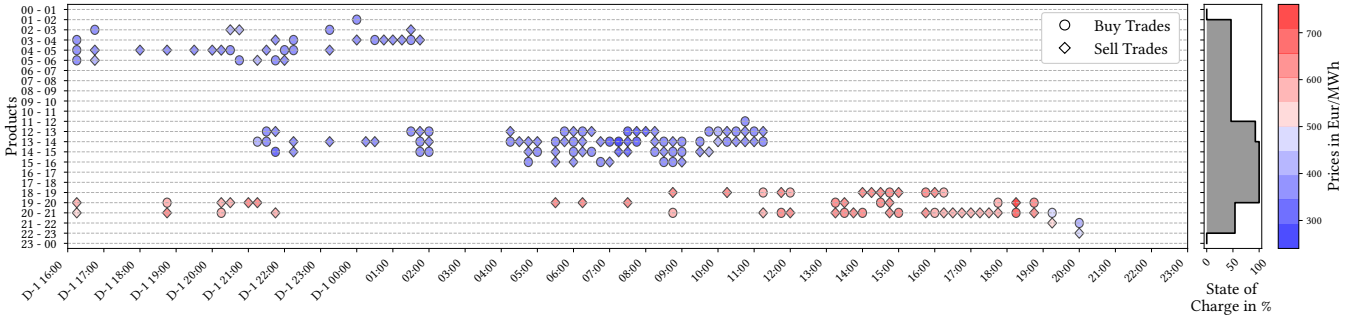

Fig. 3. Trade executions per product of the rolling intrinsic algorithm for 2022-09-07

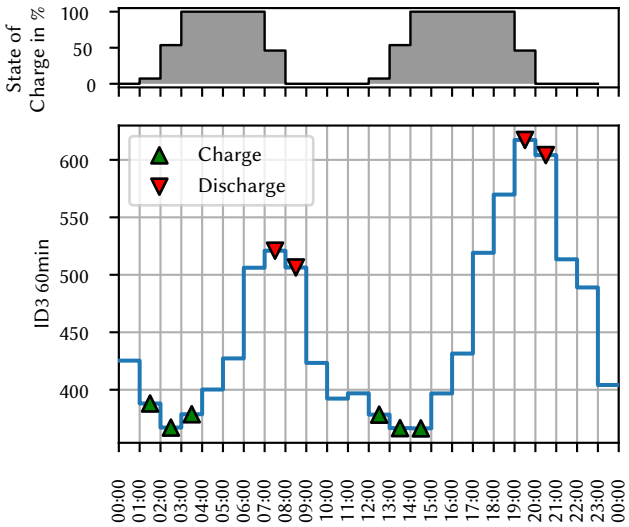

Fig. 4. Overview over the ID3 optimization results for 2022-09-07

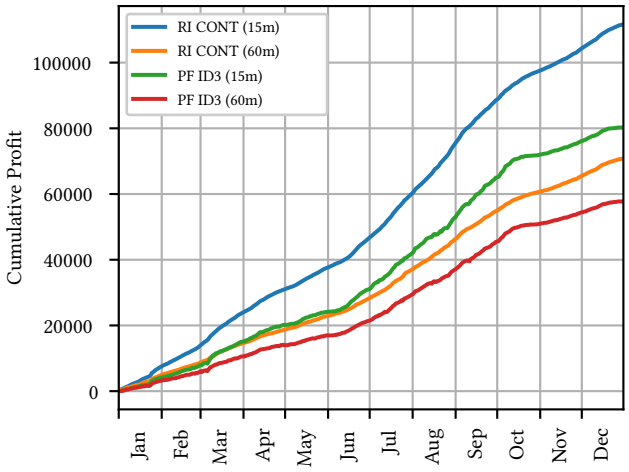

Fig. 5. Cumulative profits by trading model against full-foresight benchmarks for 2022

minimal profit made per trade, which consequently should be above the assumed transaction costs. We plan to implement a detailed profitability analysis in further studies.

Our rolling intrinsic algorithm depicts the market's reality more realistically since trades can be conducted continuously. Hence, an ID3 index-based simulation underestimated the real profit potential of battery storage systems active on the intraday market. For a profitability assessment, the continuity of the intraday markets should, therefore, be considered.

We note that our approach has a couple of limitations. First, the consideration of transaction data allows a realistic simulation based on continuously conducted trades, while keeping the computational complexity in an acceptable range. However, the whole order book data should be considered for a completely reliable analysis of profits. In addition, our algorithm assumes that a generated VWAP with 10+ trades ensures that our trade at the same price level would also be matched. Considering the low liquidity of products, even with 10+ trades and our own market impacts, this assumption might not hold in real-world settings. Especially for products with a high lead time as the price impact increases there significantly [Löhndorf and Wozabal 2022]. Second, our study only considers one single market. Previous studies have shown that considering sequential bidding on multiple markets, for instance, the reserve market, day-ahead, and intraday market, could further increase battery profitability [Miskiw et al. 2023]. We publish our source code open-source to enable a more in-depth analysis of battery systems active on the continuous intraday market in future studies, such as implementing a multi-market bidding process. Fourth, we have only benchmarked our approach against a simple ID3 simulation. Further studies should compare the algorithm against alternative methods, e.g. a Markov Decision Process. We also note that our study does not offer a detailed discussion of battery profitability on the intraday market, since we focus on the introduction of the underlying algorithm. However, future studies should employ the proposed rolling intrinsic intraday trading simulation to analyze battery-based trading profits on European intraday markets realistically.

7 CONCLUSION

Our study introduces a novel rolling intrinsic trading algorithm for battery-based trading on the continuous intraday market, leveraging a discretization method with subsequent step-wise optimization. The rolling intrinsic algorithm corresponds to the reality of the intraday market, with continuous price updates in the order book, and the thereby arising trading opportunities. In particular, the rolling intrinsic trading algorithm enables buying counterpositions, thereby realizing profits without the need for battery utilization. We show that the frequently used ID3 index-based simulation approach underestimates the profit potential of batteries active on the intraday market. We publish our algorithm open-source to enable a thorough analysis of battery trading on the intraday market in future studies. Future studies should consider the rolling intrinsic intraday market trading algorithm in combination with other markets in a multi-market bidding problem, to underline the profit potential of battery storage systems active on the power market, thereby contributing to the clean energy transition. Furthermore, a detailed analysis of BESS intraday trading profitability, considering BESS and trading costs, spreads and market access fees, has to be conducted in further studies.

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APPENDIX

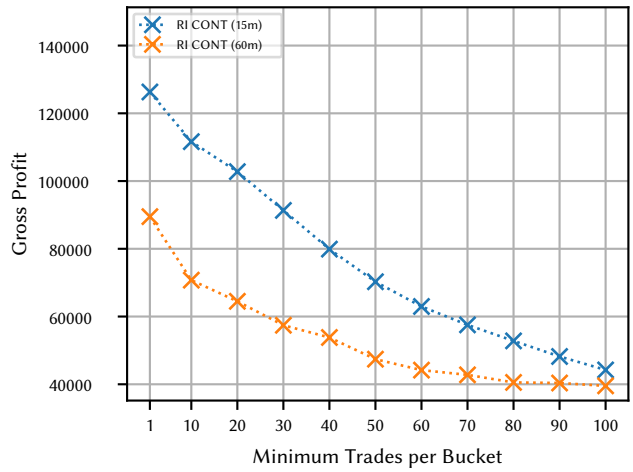

Fig. 6. Sensitivity analysis: Minimum Trades per Bucket (2022, DE/LU, 0.86% RTE, 365 yearly cycles, 0.5C C-Rate, 15m Bucket Size for RI)

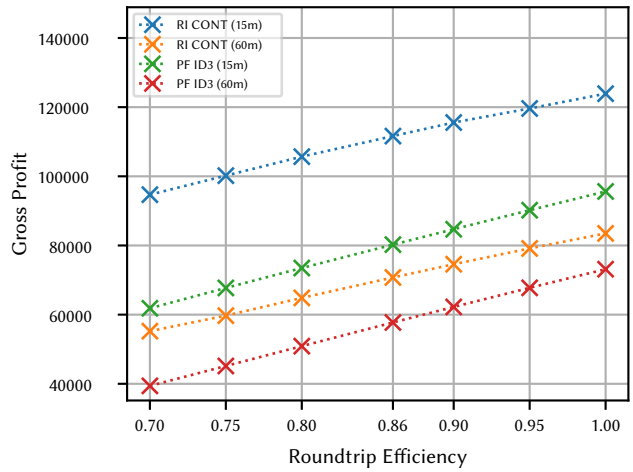

Fig. 7. Sensitivity analysis: Roundtrip efficiency (2022, DE/LU, 365 yearly cycles, 0.5C C-Rate, 15m Bucket Size for RI)

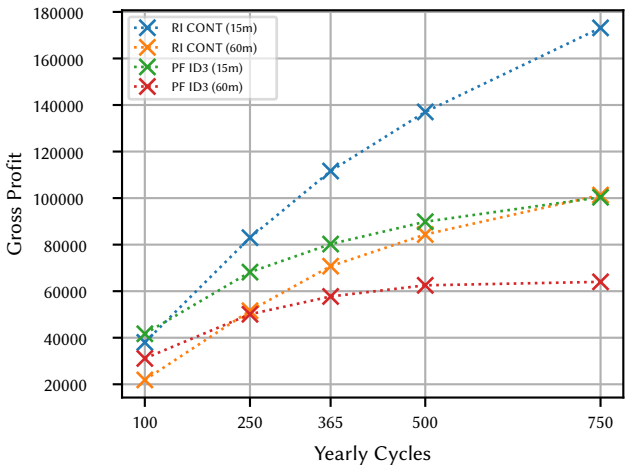

Fig. 8. Sensitivity analysis: Yearly cycles (2022, DE/LU, 86% RTE, 0.5C C-Rate, 15m Bucket Size for RI)

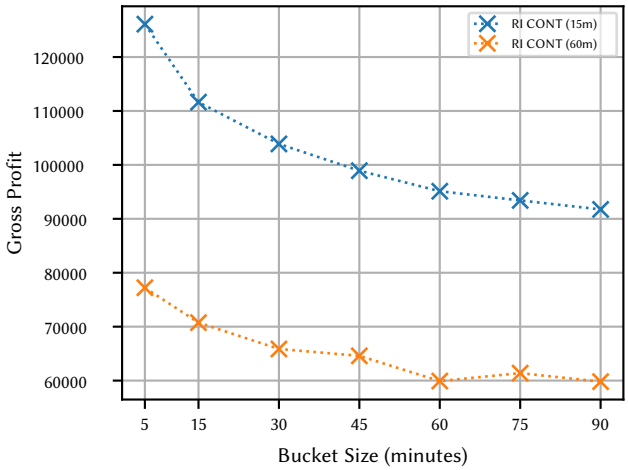

Fig. 10. Sensitivity analysis: Bucket size (2022, DE/LU, 365 yearly cycles, 86% RTE, 0.5C C-Rate)

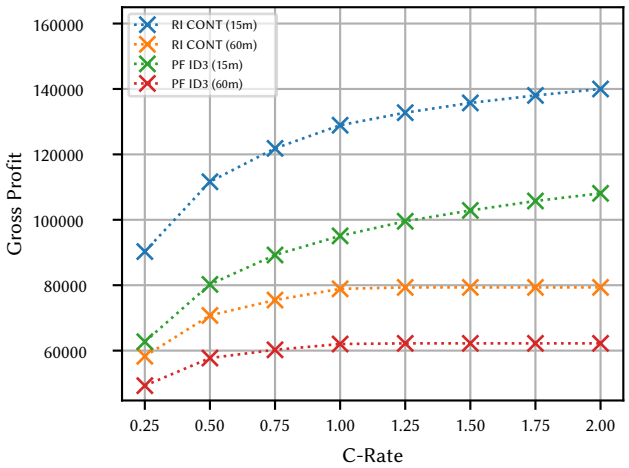

Fig. 9. Sensitivity analysis: C-Rate (2022, DE/LU, 365 yearly cycles, 86% RTE, 15m Bucket Size for RI)